

Yash Raj Singh

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Technical Skills

Java, Python, C++, SQL, HTML/CSS, JavaScript, React.js, Redux Toolkit, Node.js, Express.js, RESTful APIs, MongoDB, MySQL, JWT, RBAC, AWS, Git, GitHub, VS Code

Projects

ShopEZ <i>MongoDB, Express.js, React.js, Node.js, Redux Toolkit</i>	Jan 2025 – Apr 2025
- Engineered a full-stack e-commerce platform on the MERN stack, exposing RESTful APIs for real-time client-server communication and modular feature delivery. - Implemented JWT authentication with role-based access control (RBAC), hardening authorization paths and preventing unauthorized access during QA and UAT. - Optimized MongoDB schema design, aggregations, and indexes, reducing high-latency queries and improving response times by up to 35% under test load.	
Supply Chain Management <i>Blockchain, Smart Contracts, Distributed Ledger</i>	Aug 2024 – Sep 2024
- Developed a blockchain-backed supply-chain tracker providing end-to-end transparency, traceability, and tamper-evident event logs across stakeholders. - Authored smart contracts to verify product provenance and automate state transitions, enabling real-time origin and movement validation across the network. - Reduced fraud and reconciliation errors by 40% through immutable ledger records and cryptographic verification; improved compliance readiness via auditable transaction trails.	
Stock Market Prediction Classification Model <i>Python, ML, Reinforcement Learning</i>	May 2024 – Jun 2024
- Built a classification pipeline for stock-trend prediction achieving 92% accuracy using reinforcement-learning signals combined with ensemble modeling. - Engineered features from technical indicators—Moving Averages (MA), Relative Strength Index (RSI), and Rate of Change (ROC)—to generate actionable buy/sell signals. - Fine-tuned a Random Forest Classifier with hyperparameter search and cross-validation, improving decision boundaries and reducing false positives by ~25% versus baseline. - Documented methodology, model architecture, feature engineering, and evaluation metrics; visualized actual vs. predicted trends for technical presentation.	

Extra-Curricular

- Tech Lead**, Rajasthan Club — Orchestrated technical workshops and hackathons, leading a 15-member team; attracted 500+ participants and strengthened the club's technical brand.
- Microsoft Club — Contributed to event planning, logistics for 200+ attendees, and on-ground operations with a 12-member team to ensure seamless execution and high engagement.
- Nature and Trekking Club — Participated in environmental drives and organized trekking expeditions promoting sustainability and teamwork.

Certifications

• Oracle Cloud Infrastructure Certified Data Science Professional	Oct 2025
• Oracle Cloud Infrastructure Certified Foundations Associate	Oct 2025
• Full Stack Developer (MERN) — Smart Bridge	Jan 2025 – Apr 2025
• MongoDB Atlas & Node.js	Apr 2025
• Mastering Data Structures & Algorithms in Java — FacePrep	Jan 2025 – Apr 2025

Education

Vellore Institute of Technology, Bhopal <i>B.Tech in Computer Science and Engineering; Minor in E-Commerce Technology</i>	Aug 2022 – Present CGPA: 8.50
SOS Hermann Gmeiner School, Varanasi <i>CBSE — Class XII</i>	2021 89.6%
SOS Hermann Gmeiner School, Varanasi <i>CBSE — Class X</i>	2019 88.4%